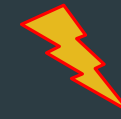


Golden Bear Circuits

Company X



Video Acceleration via Graphics CoProcessor



Hardware .v. Software Tradeoffs

± General trends & impact on design ±

Software...

Convenience

- Existing Company-X codebase/libraries
- Rapid & familiar development "flow"

Optimization Impacts

- Non-intuitive/fragile code tricks
- Distract from primary focus
- Complicate validation tests

Timeslicing to address Latency

- Inherently complex
- Overhead (context switching)
- Difficult to guarantee (non RTOS)

Hardware...

- Dramatic boosts possible
- CPU offload/parallelism

- Offload to perf/parallel specialists
- Modular separation possible

Performance

Design/Test Effort